

## WEEK 4 LABORATORY — PUTTING THE TABLES TOGETHER

*Merging queries · orphan keys · refresh*

|                  |                                                                  |
|------------------|------------------------------------------------------------------|
| <b>Session</b>   | Saturday, 5 September 2026 · four hours, in person               |
| <b>Outcome</b>   | Combine multiple business data files and explain the ETL process |
| <b>Milestone</b> | B2 — Cleaned and Documented Dataset (Weeks 3–4)                  |
| <b>Due</b>       | Monday, 7 September 2026, 11:59 PM                               |
| <b>Submit to</b> | b2-cleaned-dataset in BDAT04-2026_T1                             |
| <b>Next week</b> | Preliminary Assessment, Saturday 12 September                    |

**One sentence to carry all day.** Append adds rows. Merge adds columns. Everything in this laboratory is one of those two things, and every mistake you will make today is doing one when you meant the other.

### 0. Rebuild before you join

You cannot join tables that are not there. Every one of you has something to repair from last week before this laboratory can start.

1. Open your model and go to Power Query.
2. Delete every step you added in order to *look* at something — Group By, Keep Duplicates, Keep Errors, and any filter you added to see a count.
3. Keep only the changes you decided on. Rename each one so it says what it does.
4. Set the data type of every column you will join on. **A text key will not reliably match a numeric one.**
5. Refresh. Confirm your row count is what you expect before you go further.

**Expected after a correct rebuild:** tblSales 10,566 rows and 9 columns · tblInventory 1,541 rows and 6 columns · tblEmployees 33 rows. If your numbers differ, find out why before you merge anything.

### 1. Append — the July file

Operations has sent through July. It is a separate file, **KapeTayo\_Sales\_Jul2026.xlsx**, the same shape as your sales table with a new month in it. This is how a warehouse actually receives data: not once, but every month, forever.

#### 1a. Load it

1. Get data → Excel workbook → **KapeTayo\_Sales\_Jul2026.xlsx** → select **tblSalesJul2026**.
2. Record the row count before you do anything else: **601**

#### 1b. Append it

1. Select your cleaned **tblSales** query.
2. Home → **Append queries** → choose the July query → OK.
3. Record the new row count: **1202** Does it equal the two counts added together? **Yes**

**Now look at your columns. Count them.**

You had nine. How many do you have now? **10**

Something has gone wrong, and it has gone wrong silently. Find it before you read on — the answer is in the column headers, not in the rows.

- I did not notice anything wrong with the columns.

### 1c. What happened

Append matches columns **by name**. Any name that does not match on both sides becomes its own column, half full of nulls. Two columns now hold the same fact and neither one is complete.

1. Undo the append.
2. In the July query, rename the offending column so it matches yours exactly. Name the step: **Renamed to match tblSales**.
3. Append again. Confirm you are back to nine columns and the rows still add up.

**Write one sentence for your transformation notes explaining why you renamed it rather than deleting either column.**

- I renamed the column to match the existing tblSales column because both columns contain the same information, so deleting either one could remove valid data.

## 2. Merge — and the question it answers

A merge attaches columns from another table where a key matches. The useful part is not what matches. It is what does not.

### 2a. Join kinds

| Join kind   | What it keeps                        | Use              |
|-------------|--------------------------------------|------------------|
| Left outer  | All of mine; theirs where it matches | The normal join  |
| Inner       | Only rows that matched on both sides | Handle with care |
| Left anti   | ONLY my rows that found no match     | Your diagnostic  |
| Full outer  | Everything from both sides           | Rare             |
| Right outer | All of theirs                        | Rare             |
| Right anti  | Only their unmatched rows            | Rare             |

**Inner join is the dangerous one.** It deletes every row that did not match and says nothing at all. If you use it, you must be able to state how many rows it removed — which means running the anti join first.

### 2b. The move of the day — left anti join

1. Home → **Merge queries** → as new.
2. Pick your two tables and click the joining column in each.
3. Join kind → **Left anti (rows only in first)** → OK.
4. What comes back is every row whose key does not exist in the other table. Count it.

**Last week you asked: how many values are in this column? This week: how many of them are real?**

- I found 52 sales rows.

## 3. The integrity lab

Six joins. Predict each one before you run it. One of the six comes back empty, and that is the most important result on the page.

| Relationship               | On         | Predict | Orphan rows found |
|----------------------------|------------|---------|-------------------|
| tblSales → tblBranches     | BranchID   | 50      | 0                 |
| tblSales → tblProducts     | ProductID  | 10      | 0                 |
| tblSales → tblCalendar     | Date       | 365     | 1202              |
| tblSales → tblCustomers    | CustomerID | 500     | 560               |
| tblInventory → tblCalendar | Date       | 100     | 82                |
| tblEmployees → tblBranches | BranchID   | 50      | 1                 |

For each one that is not zero, write down the distinct offending values, not just the count. A number tells you how big the problem is; the values tell you what it is.

### 3a. The one that comes back empty

One relationship has 4,775 blanks and zero orphans. Write down which one, and then write two sentences on the difference between these:

| Blank                                                      | Orphan                                                                |
|------------------------------------------------------------|-----------------------------------------------------------------------|
| No value was recorded at all.                              | A value was recorded, and it points at something that does not exist. |
| A walk-in customer who never joined the loyalty programme. | A sale at a branch that is not in the branch table.                   |
| A fact about the business.                                 | A fault in the data.                                                  |

*Missing is not broken. A student who deletes blanks because they look untidy has just removed 45% of this company's revenue.*

## 4. What you do with an orphan

Four options. None of them is deleting it quietly.

|          | When                                                         | Example in this case                   |
|----------|--------------------------------------------------------------|----------------------------------------|
| FIX      | The value is a recognisable typo for something real          | b01 is obviously B01                   |
| EXCLUDE  | It genuinely is not part of the analysis                     | Report the count you removed           |
| ESCALATE | You cannot tell, and guessing would be inventing data        | P99 sold nine times and does not exist |
| ACCEPT   | The value is real; the reference table is what is incomplete | B07 opened in July                     |

**Whichever you choose, it goes in the step name and in your notes.** A silent fix is the offence — not the fix itself. Somebody reading your query in November has to be able to see the decision and disagree with it.

**Two of the orphans you will find today are not faults at all. Work out which two before you touch anything.**

The two orphans that are not faults are:

B07 in Employees → Branches

- Accept because the branch table is incomplete.

Missing dates in the Calendar table

- Accept because the dates can still be valid, but the Calendar table is incomplete.

**FIX:** B-02, B-05, B03, B06, b01, b04

**ESCALATE:** P99

## 5. Parameters — stop typing the file path

If your query contains a path like `C:\Users\yourname\Downloads\...` then your model works on exactly one machine and can never refresh anywhere else. Fix it now; it takes four minutes.

1. Home → **Manage parameters** → New.
2. Name it **SourcePath**, type Text, and paste your folder path as the current value.
3. Open each query's Source step and replace the typed path with the parameter.
4. Change the parameter once and every query follows.

**This is the difference between a file that works on your laptop and a model that works.**

## 6. Query folding — in concept

Where it can, Power Query pushes your steps back to the source and lets the source do the work. Filter early and the source filters for you. Add a custom column first and it cannot — so every row is dragged across before anything is thrown away.

**The rule for this course:** remove rows and columns as early as you can; add clever things as late as you can. Step order is not only about readability. On a real dataset it is the difference between four seconds and four minutes.

## 7. The refresh

The course gate, from the syllabus: your model must refresh end to end from the supplied source files **without manual repair**. A report that cannot be refreshed is not a report — it is a spreadsheet that happened once.

1. Close and apply. Refresh the whole model from source.
2. Every query must complete. If any one fails, the milestone is not finished.
3. Then export your model: Model view → TMDL View → drag each table in → copy → save as **model\_2026-09-05.tmdl** in your backups folder.

## 8. Two decisions, out loud, no notes

Before you leave, each of you will stand up for three minutes and defend two cleaning decisions from your own work. This is not a presentation and there are no slides.

| Say                      | Meaning                                          |
|--------------------------|--------------------------------------------------|
| What you did             | Name the step and how many rows it touched       |
| Why you were entitled to | The business reason, not the technical one       |
| What was lost            | Every cleaning step destroys something. Name it. |

**Next Saturday's preliminary assessment carries this as a graded criterion. Today is your rehearsal, and it is free.**

## B2 — what to submit

| Item                        | Detail                                                                  |
|-----------------------------|-------------------------------------------------------------------------|
| Appended and merged queries | The case tables joined into an analysis-ready set                       |
| Transformation notes        | Every decision, its reason, and what it discarded                       |
| Integrity report            | Six relationships tested, orphans counted, a decision recorded for each |
| A successful full refresh   | End to end from source, no manual repair                                |
| TMDL export                 | model_2026-09-05.tmdl in your backups folder                            |

|                                                                                                                                                                                                                                 |                            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| AI Use Log                                                                                                                                                                                                                      | Updated. Tier and purpose. |
| <b>Build B2 on the ORIGINAL case file.</b> Whatever file arrives during the session is for the session only — re-point your source before you leave the room, or you will submit a model that refers to a file nobody else has. |                            |

## If you get stuck

*Fifteen minutes, then ask. You are working in pairs today, so ask your partner first and me second. Tell me the section number, what you clicked, and what happened.*

— Sir JB